

TLC and Column Chromatography

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Abstract

The objectives for this TLC (Thin Layer Chromatography) experiment were to answer the questions: what eluent would best produce good lab results, how to calculate R_f (retention factor) value for pure and mixture samples and its relationships, and how to identify certain components of a spotting (fractional) via TLC analysis? In general, we utilized standard TLC procedures to carry out several spotting analyses to answer our objective questions split into three separate parts of experiment. In our experiment, we determined the best eluent for subsequent reactions, calculated appropriate R_f values and determined what high versus low data meant, and identified components of fractions with a reference. Overall, it was concluded that 1:1 petroleum ether:diethyl ether was the best eluent for our TLC experiment, lower R_f values correspond to more polar compounds due to a higher attraction to the stationary phase (vice versa), and we correctly identified acetylferrocene and ferrocene components through several fractional TLC spotting.

Introduction

This report encloses the importance of TLC chromatography in chemistry when determining which compounds are within a mixture. Chromatography is a general method of separation where solutes are separated between the stationary (polar-like) and mobile (nonpolar-like) phases. In TLC (thin layer chromatography), the stationary phase is normally either silica or alumina powder absorbed on a glass backing (solid plate), and the mobile phase involves a specified liquid solvent (eluent). During the experiment, the TLC plate is placed in the eluent to develop itself. In general, the more polar a compound is, the more likely it will stay absorbed to the stationary phase, while the less polar a compound is the more likely it will dissolve in the mobile phase and be carried to the top of the TLC (called the solvent front). Since polar compounds generally do not move much past the starting line compared to their nonpolar counterparts, it will generally have a small R_f value (discussed below) while nonpolar compounds will have a larger R_f value. If many of them are separated near the bottom, it is best to use a more polar eluent so that polar compounds travel farther from the starting line, which would make the nonpolar compounds move slightly downward away from the solvent front. Finally, this technique can be used to identify which compounds are within a certain unknown mixture of known material when comparing to a known R_f value of said material – known as TLC analysis.

Another method that can be used for mixture separation is column chromatography (experiment not executed in this report). For this type of separation, the stationary phase is packed into a vertical column while the mobile phase (a gel-like structure) is pulled through the column via gravity. The theory behind polar and nonpolar separation between both phases is like TLC, but polar compounds will remain towards the top of the column while nonpolar compounds will be toward the bottom of the column due to being pushed by the mobile phase through gravity. Finally, a graphic of this technique is visualized in *Figure 1*.

In general, TLC is most often used for identifying or monitoring reactions through analysis, while column chromatography is used to separate components to isolate the desired pure product through isolation.

Furthermore, R_f (retention factor) values are a way to compare and measure how attracted a compound is to the stationary phase and was utilized through this entire experiment (equation in later sections). In addition, two factors affect the value of R_f (the polarity of compound and polarity of eluent). First, the greater the polarity of the compound, the lower its R_f value because it is very attracted to the stationary phase. Second, the greater the polarity of the eluent, the higher the R_f value for polar compounds and

lower R_f value for nonpolar compounds (factor utilized in this experiment), vice versa. Finally, it is important to note that this theory was tested in our experiment, and we determined its legitimacy.

The purpose of this experiment was to identify which eluent would better receive good results, find and understand the R_f value for several spot separations, and identify which compounds were separated in several fractions through TLC analysis. In short, during the experiment, we took several TLC trials to discover and understand the results and theory behind this separational analysis via the three objectives mentioned above.

For this experiment, I hypothesize that the appropriate eluent will be an even mixture between polar and nonpolar compounds. Furthermore, I expect that the polar compound will have a lower R_f value than the nonpolar compound. Finally, I anticipate that all the compounds will separate appropriately so that they can be identified by their R_f values.

Ultimately, it is important to note that this report is outlined into three parts. Part I demonstrates how we determined which eluent was appropriate, part II demonstrated how we calculated and understood R_f values, and part III demonstrated how we identified several compounds based on their affinity for stationary/mobile phase within four TLC fractions.

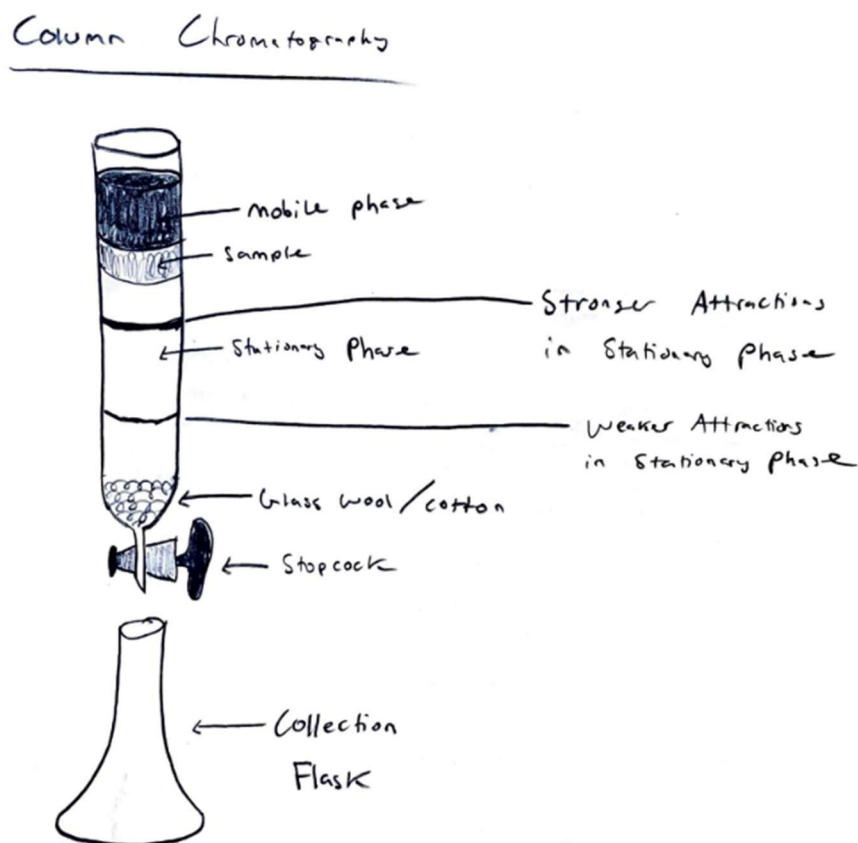


Figure 1: Column Chromatography

Procedure

The procedure of this experiment entails three portions, and they are as follows:

Material

<u>Equipment</u>	<u>Chemicals</u>
▪ TLC Plates and spotters ▪ Pencil ▪ UV lamp ▪ Scotch tape ▪ Pipette	▪ Mixture of acetylferrocene & ferrocene ▪ Petroleum ether ▪ Diethyl ether ▪ 9:1 petroleum ether/diethyl ether ▪ 1:1 petroleum ether/diethyl ether

Equations

$$R_f = \frac{\text{Distance traveled by substance}}{\text{Distance traveled by solvent front}}$$

Methods

The general methodology for preparing this TLC experiment is described below. We began by drawing a pencil line about 1.00 cm from the bottom of the prepared TLC plate. Next, we used a pipette to spot the TLC with only a small amount of material – about 1.00 mm in diameter once placed and allow to air dry. Then, we placed a filter paper inserted into a beaker that is filled with eluent (hold the plate upright outside the beaker so that the level of solvent in beaker is below the pencil line). After, we placed the plate in the beaker (the pencil side down) so that the solvent moves up the plate. When the solvent edge had moved up about 75% of the TLC plate, we removed it and pencil drew the solvent front before it evaporated. Finally, after we allowed the solvent to evaporate, the TLC was visualized with a UV lamp and circled to identify their location and measured their distance from the starting line. Overall, this technique was utilized for all three parts demonstrated below.

For part I, three different solvents (100% petroleum ether, 9:1 petroleum ether:diethyl ether, and 1:1 petroleum ether:diethyl ether) were used individually on ferrocene and acetylferrocene. After the experiment, we determined which eluent would best work for this experiment and R_f values were determined for each sample per solvent.

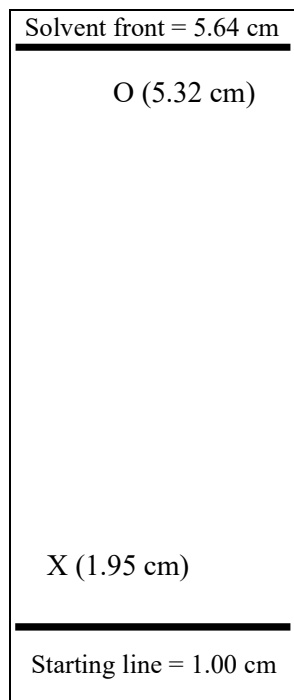
For part II, the chosen eluent was used for three TLC plates (mixture, pure ferrocene, and pure acetylferrocene) and R_f values were determined for each separated spot per plate.

For part III, four TLC fractions were used, instead of column chromatography, utilizing the same eluent for an unknown, single spotting. Then, each spot separated into its single or multiple spots per fraction and R_f values were calculated for each.

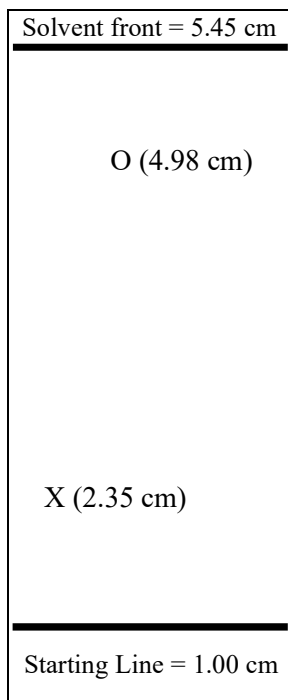
Results

Part I:

100% Petroleum Ether



9:1 Petroleum Ether:Diethyl Ether



1:1 Petroleum Ether:Diethyl Ether

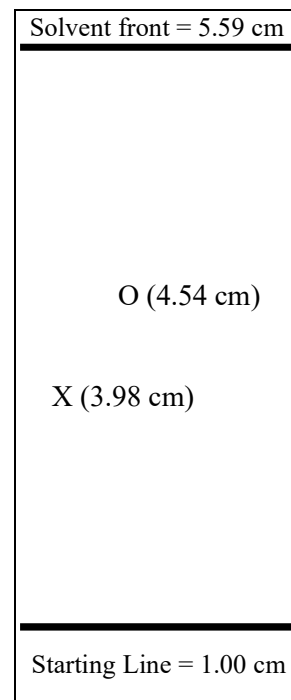


Figure 2: Initial TLC Analysis – Determining the Solvent

Key: X and O represent acetylferrocene and ferrocene, respectively.

Table 1: R_f Values for Initial TLC Analysis

	100% Petroleum Ether	9:1 Petroleum Ether:Diethyl Ether	1:1 Petroleum Ether:Diethyl Ether
Ferrocene (R_f)	0.931	0.894	0.771
Acetylferrocene (R_f)	0.205	0.303	0.649

Calculation 1: Example Math – Finding R_f for Ferrocene in 100% Petroleum Ether Eluent

$$\Rightarrow R_f = \frac{\text{Distance traveled by substance}}{\text{Distance traveled by solvent front}}$$

- Distance Traveled by substance = 5.32 cm – 1.00 cm = 4.32 cm
- Distance Traveled by solvent front = 5.64 cm – 1.00 cm = 4.64 cm
- $R_f = \frac{4.32 \text{ cm}}{4.64 \text{ cm}} = 0.931$

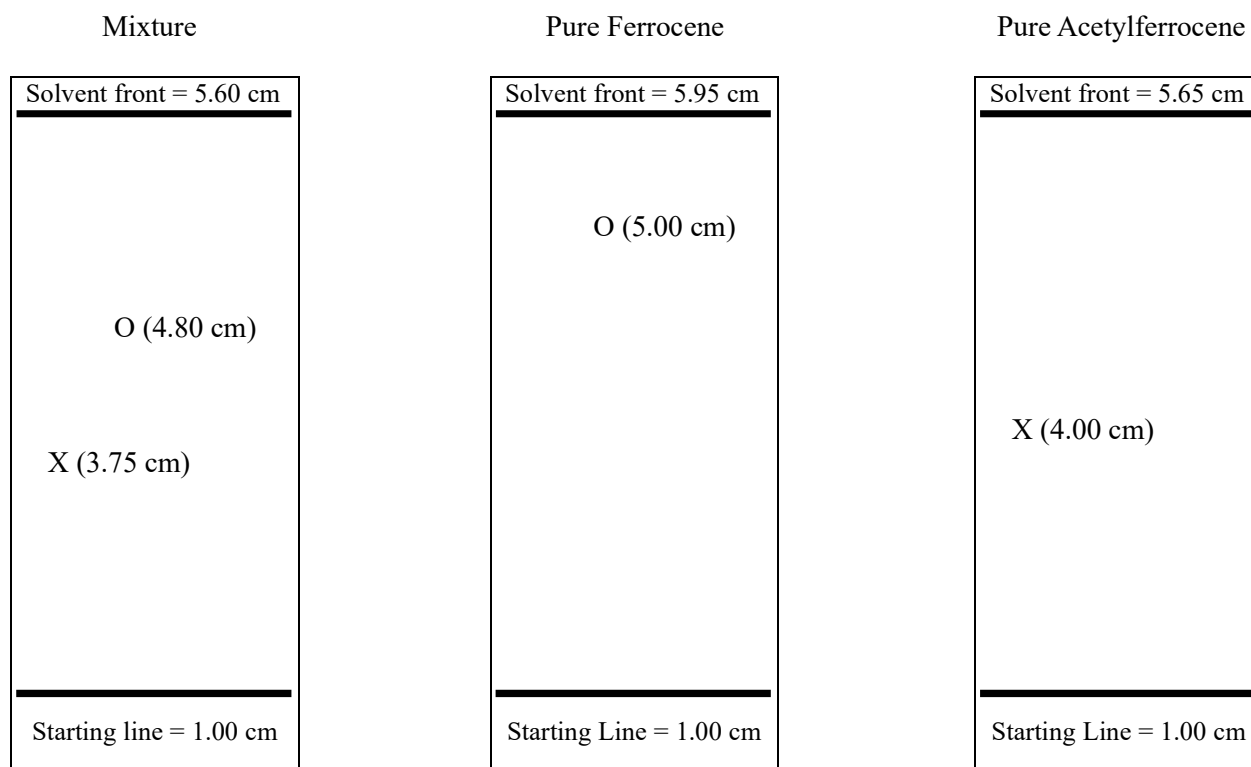
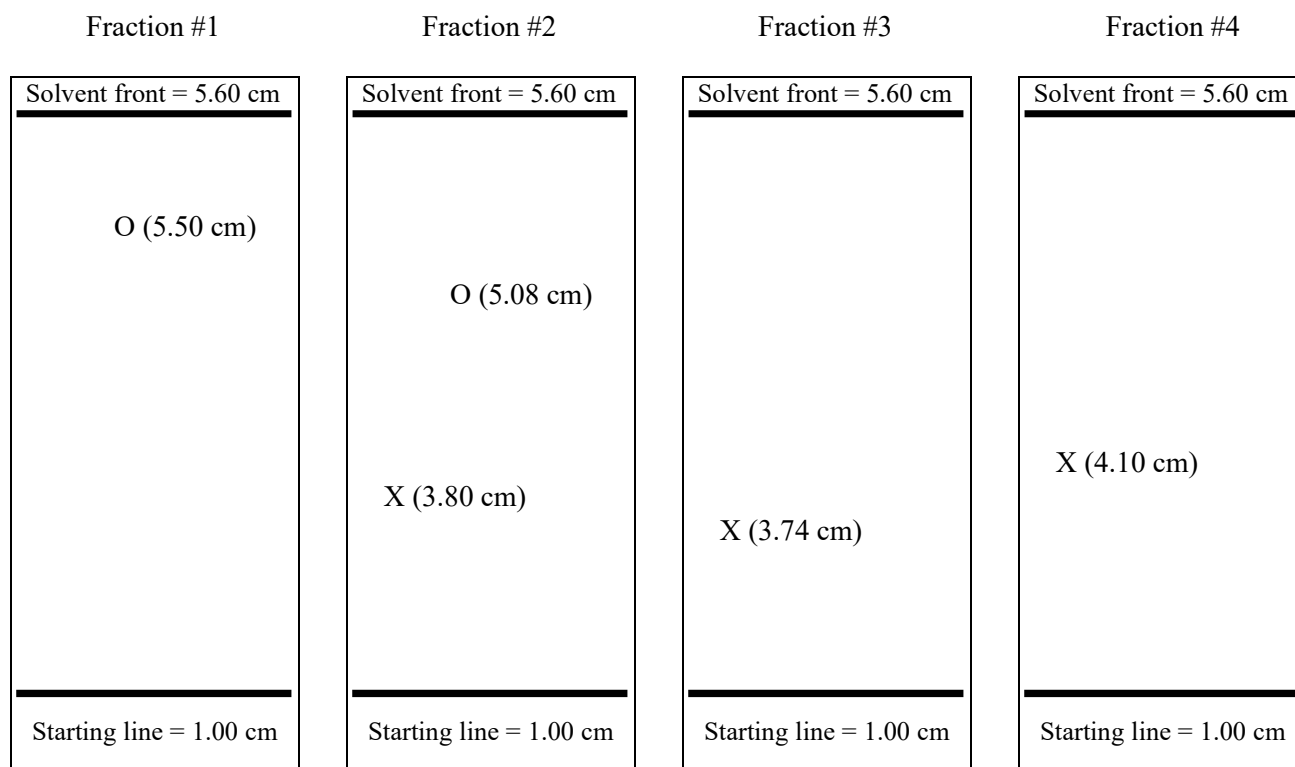
Part II:

Figure 3: Comparing R_f Values from Pure and Mixture Substances

Key: X and O represent acetylferrocene and ferrocene, respectively.

Table 2: R_f Values relating to Figure 3

	Mixture	Pure Ferrocene	Pure Acetylferrocene
Top Spot (R_f)	0.826	0.870	0.652
Bottom Spot (R_f)	0.598	N/A	N/A

Part III:*Figure 4: Identifying Substances Through Fractions*

Key: X and O represent acetylferrocene and ferrocene, respectively.

Table 3: R_f Values relating to Figure 4

	Top Spot: R_f	Top Spot: Identity	Bottom Spot: R_f	Bottom Spot: Identity
Fraction #1	0.891	Ferrocene	0.648	Acetylferrocene
Fraction #2	0.887	Ferrocene	0.609	Acetylferrocene
Fraction #3	0.596	Acetylferrocene	N/A	N/A
Fraction #4	0.674	Acetylferrocene	N/A	N/A

Discussion

Part I:

This section demonstrates how we determined which eluent would better produce good data for the remainder of the experiment. *Figure 2* gives a visual representation of the TLC plates, which signaled that as a higher concentration of polar solvent was in the eluent mixture, the polar compound (acetylferrocene) traveled farther up from the starting line, giving better lab results – even though the nonpolar compound shifted slightly downward (ferrocene). *Table 1* represents the R_f value comparisons (with example calculation in *Calculation 1*) to support these visual findings. It can be seen from the table that the R_f value of acetylferrocene increased as the polar concentration in eluent increased, while only slightly decreasing the ferrocene R_f value. Therefore, the 1:1 petroleum ether:diethyl ether was the best eluent for this experiment compared to 100% petroleum ether and 9:1 petroleum ether:diethyl ether. Finally, it can be said that the polarity of the eluent influences the R_f value.

Overall, these results agree with my hypothesis that the more even mixture between polar and nonpolar material in eluent will make better lab results as the R_f value for the polar compound will be more acceptable.

Part II:

This section determined how the polarity of different pure compounds affects the R_f value/affinity of the substance to the stationary/mobile phase. *Figure 3* gives a visual representation of the TLC plates that were utilized to compare R_f values. Additionally, the two separations of the mixture plate were identified utilizing the single separation per pure plate and relating their R_f values – all using the determined eluent from part I. When analyzing the data from *Table 2*, it can be said that the polar compound (acetylferrocene) tended to have a lower R_f value (0.598 – 0.652) compared to the nonpolar compound (ferrocene) with higher R_f values (0.826 – 0.870). As mentioned, lower R_f values correspond to a compound with higher affinity to the stationary phase, vice versa. Therefore, acetylferrocene, a polar compound, has a higher affinity for the stationary (polar) phase, while ferrocene, a nonpolar compound, has a higher affinity for the mobile (nonpolar) phase. All in all, it is proven that the polarity of the sample influences the R_f value.

Overall, these results agree with my hypothesis that the lower R_f value compounds will be more polar than the R_f value of the nonpolar compounds.

Part III:

This section analyzed the separation of four fractions and compared their R_f values with known values (previous parts) of acetylferrocene and ferrocene to determine which compound were within each spotting. It is important to note that this experiment was done for us through column chromatography, so we analyzed it through TLC. *Figure 4* gives a visual representation of the TLC plates that were utilized to conduct our analysis while *Table 3* demonstrates computational data and spot identification. Finally, it was determined that R_f values consisting of 0.800+ were considered ferrocene when comparing to previous parts of this experiment, while R_f values consisting of 0.590 – 0.680 were considered acetylferrocene utilizing similar analysis techniques.

Overall, these results agree with my hypothesis that the two compounds will separate appropriately within their fractions to accurately compare their R_f values and determine their identity.

General Discussion

Regarding potential errors that could have given slightly unexpected results, several reasons could have been at play. First, hand oils could have seeped onto the TLC plate creating an impure mixture of sample. Second, portions of the TLC plate could have touched the table or pipette resulting in an unsanitary plate or impure mixture, which would skew results. Finally, the pure substances or mixtures could have been contaminated, which would skew the affinity of each sample to the stationary/mobile phase.

Conclusion

In summary, this experiment has succeeded in properly demonstrating that the best eluent was a well mixture between polar and nonpolar compounds (1:1 petroleum ether:diethyl ether), R_f values would be lower for more polar compounds (as they are more attracted to the stationary phase), and all fractional spotting would appropriately separate so that they could be accurately analyzed and matched to their correct components via R_f . These findings support all three of my hypotheses directly, indicating that we correctly expected what would happen during this experiment. Additionally, any possible deviation from perfect data could have been from an assortment of human errors, such as hand oils interfering, unsanitary plate/environment, and impure mixtures. For the future, I would suggest that more careful craftsmanship and an ideal environment could be utilized to get even better data. Finally, it would be advised that a column chromatography experiment would be performed by us (instead of provided) so that we could fully understand the depths of column separation instead of TLC analysis.